

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 03)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

0 1 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

5

0 2

A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 2 . 1

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer t/ha

0 2 . 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 2 . 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1

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5

0 3 . 1

For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

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Whether the numbers of dormice differ between woodlands with different management ..

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Whether mean dissolved oxygen differs between two streams

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Whether crop yield increases with fertiliser application (ranked data)

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0 3 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 3 . 3 A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 3 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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10

04

Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m ⁻² day ⁻¹	Mean solar energy in December / kWh m ⁻² day ⁻¹
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

04

. 1

Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer

times

04

. 2

Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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04

. 3

Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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04

. 4

Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

04 . 5

Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

Extra space

15

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Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

- A High sand content
- B High clay content
- C Crumb peds
- D High pH

□ □ □ □

0	5	.	2
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Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

0 5 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

5

0 6 . 1

Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

Extra space

06 . 2

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

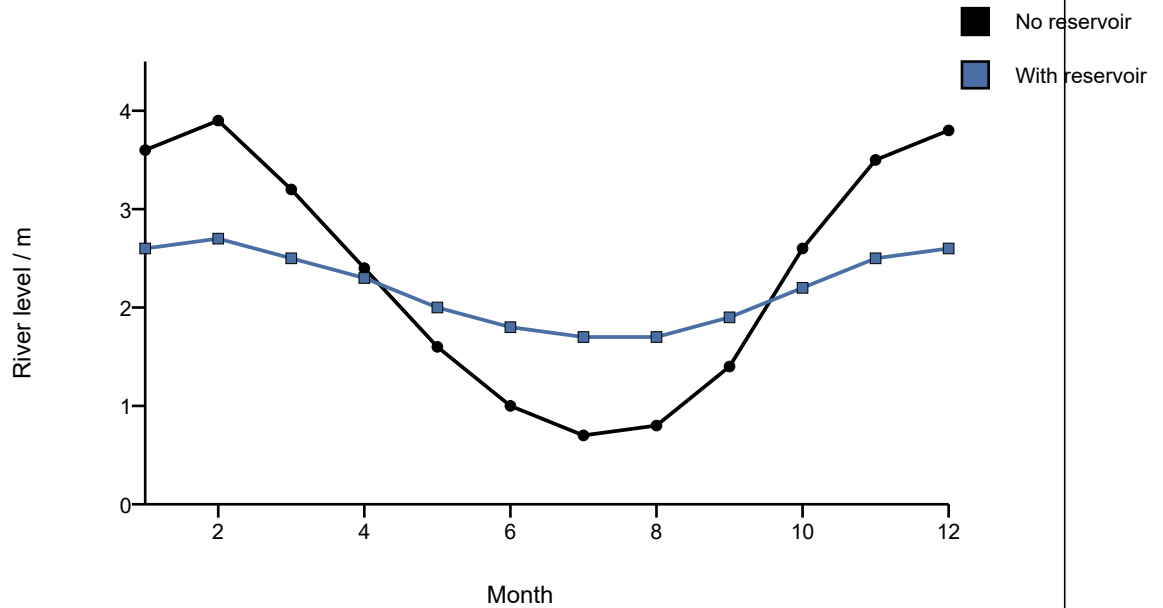
[1 mark]

10

07

Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



07

1

Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

m

07

2

Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

0	7	.	3
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Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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15

08 . 1

Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

- A Ozone is destroyed only by CFCs ☐
- B Ozone absorbs mainly infrared radiation ☐
- C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐
- D Ozone concentrations are highest at ground level ☐

08 . 2

Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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08 . 3

Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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09

Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

09 . 1

Explain the purpose of screens and grit traps.

[2 marks]

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09 . 2

Explain why primary sedimentation is described as the most important process.

[2 marks]

09 . 3

Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

09 . 4

Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

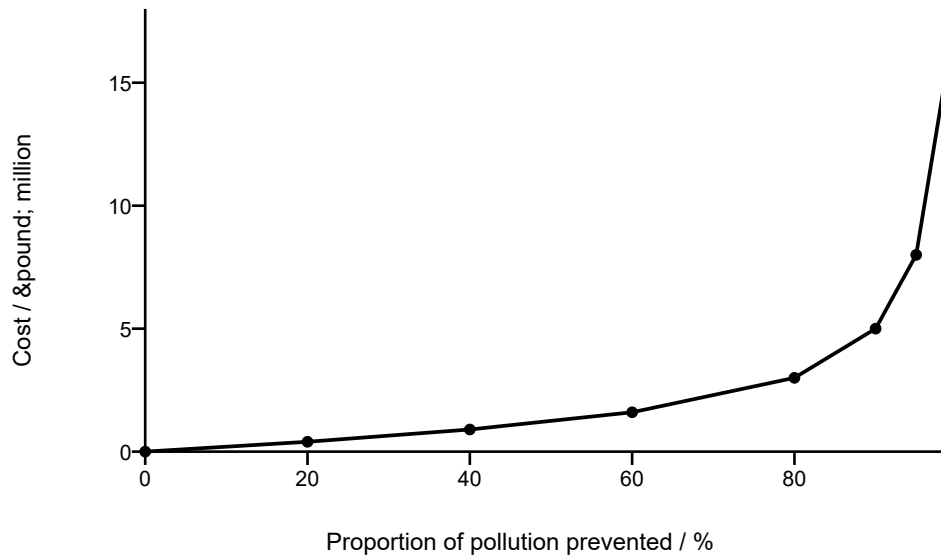
[3 marks]

10

1 0

Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0

. 1

Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer

£ million

1 0

. 2

Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

1	0	.	4
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Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1	1	.	2
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Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

This image shows a full page of primary-ruled paper. It features approximately 20 horizontal dashed lines spaced evenly down the page, providing a guide for handwriting practice. The paper is otherwise blank, with no margins or additional markings.

END OF QUESTIONS